



Unit 1: Introduction to Project Management

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Introduction to the Unit

In this introductory unit, we will learn about the basic concepts of project management. Upon completion of this unit, we will have a fair understanding of the fundamentals of project management.

We will focus on understanding the *Initiation Phase* and the various activities and processes performed in this phase. Some of the concepts that we will examine closely will include the *Project Proposal*, the *Project Charter*, the *Project Life Cycle*, the *Project Kick-Off* and the difference between *Project Management* and *Programme Management*.

Objectives

Upon completion of this unit, you will be able to:

- Describe the various functions performed in the *Project Initiation Phase*
- Explain the importance of the *Project Proposal* and the *Project Charter*
- Describe the process of identifying and selecting projects
- Explain the importance of having a project boundary
- Describe the importance of a project kick-off meeting
- Differentiate between project management and programme management
- Identify the guidelines that govern the process of portfolio management



Defining the Project

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1. Think About This
2. Essentials for Defining a Project
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Objectives

Upon completion of this segment, you will be able to:

- Describe the *Project Defining Phase*
- Identify the essential factors to be considered when defining a project

Think About This

Welcome to the first phase of the project management life cycle—the defining phase. *The Project Defining Phase* is the first phase of the project management life cycle during which a project is initiated, defined and begun.

How do you think projects are begun?

There are many reasons why a project might be undertaken.

Let us briefly discuss the case of the Eiffel Tower.

It was a project commissioned to celebrate the French Revolution and was completed in 1889.

It was built:

- to make an artistic statement to celebrate freedom and democracy,
- to remember the glorious history of the French and to forget their defeats, and
- for therapeutic reasons, unlike, say, a dam (which is built for practical use as it controls water and often provides hydroelectric power).

Basically, it was constructed as a monument dedicated to the French and their many successes. Over the years, however, the tower has come to serve many useful functions in addition to providing several excellent restaurants to the city of Paris. This contradicts its original description as ‘useless but therapeutic’.

In 1909, in particular, it was used as a radio tower. When the military intercepted strategic enemy messages during World War I, the radio tower was transformed from a temporary set up to a permanent one. Moreover, every year, the tower continues to attract millions of tourists, who come and spend a large number of Euros in the city of Paris.

Thus, as we can see, a great project can be started or initiated for one reason, but can fulfil many additional roles during its useful life, some of which might not be obvious at its inception. Success



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can be measured not only in terms of cost and magnitude of achievement in the original construction and design, but by purpose and unanticipated future uses as well.

But, think about this!

In today's context, it is difficult to envisage a major project being launched solely for symbolic reasons (as a monument to signify pride, achievement and/or accomplishment). These days, more often than not, the overriding consideration in initiating projects is founded on practical, rather than emotional reasons.

As you put yourself in the position of a *Project Manager* and embark on the *Defining Phase* of the project management life cycle, you will explore the factors to consider when initiating, defining and beginning a project.

Essentials for Defining a Project

Successful project management involves effective communication and co-ordination and the setting of overall project priorities. A lead *Project Manager* who is willing to take risks is a mediator, a negotiator, a team builder and a true leader or champion for the project.

In managing a project, we must establish what is it that we wish to accomplish, with whom and by when. We also have to be fully aware of the constraints we are up against in completing the effort.

In the defining phase, we must:

- be aware of how the project fits in with the strategic objectives or goals of the firm,
- detail what we hope to accomplish when the project is completed,
- determine what is the scope of the project,
- identify the trigger or mechanism which initiated the project,
- be aware of how projects are selected for funding (Why did one project get selected, while others did not?),
- understand the project management method of organising as opposed to some other organisational model,
- decide how to balance the competing objectives of scope, time, cost and quality, and
- identify the major stakeholders who must be considered.

These are all considerations the project manager must take into account before launching any new project. A project is often distinguished from other types of work activity in that it requires the use of many and varied resources and focuses on a unique deliverable.

Each project is unique. It is important that we identify early on in the project management process what specifically is to be accomplished when the project is completed, that is, what is the *unique deliverable*?



Summary

In this segment, you learned that:

- the *Defining Phase* is the first phase of the project management life cycle;
- during this phase, a project is initiated, defined and begun;
- projects are defined on the basis of practical considerations;
- project management requires clarity on what is to be accomplished, with whom and by when; and
- before the launch, a *Project Manager* must consider the key essentials of defining a project.



Writing the Project Proposal

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1. Introduction
2. Components of a project proposal
3. Summary

Objectives

Upon completion of this segment, you will be able to:

- Identify the components of a project proposal
- Prepare a project proposal

Introduction

In the previous segment, we learned about the project defining phase and the factors to be considered while defining a project. In this segment, we will focus on the project proposal and its components.

The project proposal is the final planning document. It is the culmination of a great deal of effort, usually involving many individuals. It is the bridge between the recognition of a problem or an opportunity and the beginning of the implementation of a solution for it.

If a project proposal has been developed wisely, the project has a much higher likelihood of being completed on time, within scope, within budget and at the quality level that all major *Stakeholders* expect. If it has not been well prepared or thought out, the project is more likely to be doomed to failure or an unsatisfactory completion. Many problems lie ahead in these instances.

Components of a Project Proposal

The items that need to be included in the project proposal vary from organisation to organisation. Most firms and government agencies desire the informational inputs we have prepared up to this point, such as the project overview, plus a somewhat more detailed description of the project.

Generally, the following are also included in a project proposal:

- a letter of transmittal (memorandum);
- project name;
- overall project budget, possibly time-phased as a graph so that the senior management is aware of the total budget and the rate of expenditure on project activities;
- the *Project Manager's* name;
- project description;
- estimated project start and finish dates;
- an identification of major milestones indicating in-process progress;
- special issues;



Summary

In this segment, you learned that:

- in order to increase the chances for success—throughout the project—it is essential to develop a detailed project proposal. Doing so will help to ensure that the project is completed on time, within scope, within budget and at the quality level all major *Stakeholders* expect; and
- the requirements for the project proposal vary from organisation to organisation. Some of the key components, however, may include (but are not limited to):
 - the project description,
 - schedule and major milestones,
 - budget,
 - the project manager's name,
 - major milestones, and
 - the WBS.



Initiating the Project

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4. Basic Steps of Project Initiation
5. Identification of Stakeholders
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Objectives

Upon completion of this segment, you will be able to:

- Describe how projects are initiated
- Explain the reasons for initiating projects
- Describe the organisational demand categories
- Describe the basic steps of project initiation
- Identify the various categories of stakeholders

Introduction

In the previous segment, we learned about the importance of the project proposal and its key components. In this segment, we will focus on understanding the various steps of the *Project Initiation Phase* and on identifying the project *Stakeholders*.

The *Project Initiation Phase* is the conceptualisation of the project. This section describes the basic processes that must be performed to get a project started. Accordingly, the purpose of this phase is to specify what the project should accomplish. The caution in this purpose is that if the customer's needs are inadequately articulated, poorly formulated goals and objectives will stand out as a significant source of concern. This starting point is critical because it is essential—for those who will deliver the product/process, for those who will use that product/process and for those who have a stake in the project—to reach agreement on the project's initiation.

Reasons for Initiating a Project

Projects are initiated when a perceived need or opportunity is seen by an individual or an organisation. Projects can be started either by an individual, society or an organisation. In an organisation, certain individuals may have the authority to start a new project; they are called the *Sponsors*. Individuals can also start projects based on their needs. Let us understand the reasons for project initiation from, both, an individual as well as an organisational perspective.



Individual

An individual can work on a project based on his/her capacity and needs. There could be many reasons for an individual to initiate a project. It could be to achieve a personal goal, convert certain ideas into action or meet the needs of his/her family or the society at large.

There are many examples of projects that can be initiated by individuals. Shifting home to be closer to the workplace could be one such project. Similarly, studying to graduate, throwing a big party for friends or tinkering with an old bike to make it as good as new could all be an individual's projects.

In such cases, the individual can be both the *Sponsor* as well as the decision maker. He/she can build a team and execute the project or do it on his/her own by performing all the roles required in that project.

Organisational

Organisations initiate projects for several reasons, which include:

- responding to a stated request from a client or fulfilling an internal need,
- solving a current or future problem,
- meeting (or creating) a market demand,
- complying with changing legal requirements or standards,
- improving a process,
- lowering the costs of an existing service,
- exploiting a technological advancement and building upon it, and
- pursuing a business strategy or an opportunity.

Organisational Demand Categories

The organisational demand for initiating projects can be broadly classified into the following categories:

Market demand

The demands of the marketplace can drive the need for a project. For example, a bank initiates a project to offer customers the ability to apply for mortgage loans over the Internet due to a drop in interest rates and an increase in demand for refinancing and new home loans.

Business demand

An organisation's business may demand that certain projects—related to its core business—be taken up. For example, construction companies often need to take up one project after another. This decision may be guided by the motivation to capitalise on the real-estate boom at any given point in time.



Customer request

Most companies have customers, and their requests can drive new projects. Customers can be internal or external to the organisation. Government agencies do not have external customers per se (we are captive customers at any rate), but there are internal customers within departments and across agencies. For example, following customer requests, a chain of fast-food restaurants may initiate a project to offer the customers value-added services such as free access to Wi-Fi and chocolate/candy vending machines.

Technological advancement

Do you happen to own one of those electronic gadgets that help store names and addresses and usually come with a calendar and a to-do list of some kind? There are always newer and better versions of these devices entering the market. The introduction of satellite communications now allows these devices to connect to the Web and access e-mail almost anywhere in the world. Therefore, satellite communications is an example of technological advancement. Keen to take advantage of this technology, electronics manufacturers routinely need to initiate projects to ensure that their products are up-to-date and competitive.

Legal requirements

The private industry and government agencies, both, generate new projects as a result of laws passed in every legislative season. For example, new sales tax laws might require new programs to upgrade the existing sales tax system. The requirement that food labels appear on every package describing the ingredients and the recommended daily allowances is another example of a project driven by legal requirements.

Social need

Project initiation may be driven by a social need. This need is a result of social demands. For example, if a developing country is experiencing an epidemic, medical supplies and facilities are needed to vaccinate and cure those infected with the disease. The relevant project for this situation will involve utilising the existing resources of the government and vaccinating everyone in the region.

Another example could be to prevent water contamination by cleaning up waste—let out of manufacturing plants—before it enters a local river or stream. Construction of dams or roads is also a good example of projects initiated to meet social needs.

An everyday example of a social need may be a talent show being organised in a college. The most likely reasons for organising such an event would be to motivate the students, showcase their talents and strengths as well as to compete with other colleges.



Basic Steps of Project Initiation

Following are the basic steps of the *Project Initiation Phase*:

Create a business case

The need for initiating a project has to be documented and discussed with all *Stakeholders*. This is done with the help of a business case document. This marks the formal start of the project when the project *Sponsor* (or the project initiator) gives a description of the business problem/opportunity. The project is initiated to address the problem or provide alternative solutions. The business case document includes the business problem and potential costs associated with the project implementation.

Perform a feasibility study

A feasibility study is a research conducted to determine whether the project can be accomplished and to identify the likelihood of alternative solutions. The feasibility study investigates whether the benefits stated in the business case can be delivered. It also depicts the relationship between business costs and project solutions.

If required, a mock-up project is done to confirm feasibility. The *Sponsor* needs to ask the following questions before beginning a project:

- What is the benefit of this project?
- Does this project resolve the current issue at hand?
- Are there any other alternatives before beginning this project?
- Are there sufficient funds available?
- Are there skilled resources available in the market?
- Can this be done within the available capability or does it need to be outsourced?
- Are there any legal issues involved?
- Does it have a market?

There are various tools and techniques available to arrive at the decision whether to start a project.

Many consulting and advisory companies do an analysis and provide inputs to organisations. They are called 'subject matters experts' or even 'consultants' in those areas of work. If a *Sponsor* does not want an outside consultant, he can identify a team comprising business analysts, market analysts and financial risk managers from within the organisation. This internal team should comprise experts in these fields who should provide him the necessary data before beginning the project. These people are called advisors or comprise the decision-support team.

A feasibility study done by experts could further confirm how the project is perceived.

Confirm terms of reference

Based on the feasibility study output, detailed goals, requirements and the manner of executing the project is determined and documented. It also includes financial details and scope at high level. A *Project Initiation Document* is created at this juncture. It will contain all the details about the project



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such as the reason for initiating, its scope, risks, assumptions, constraints, benefits, target consumers, market analysis, gap in market, time-to-market, loss analysis and so on. This document helps the business *Sponsor* or the individual decision maker. This document also serves the *Sponsor* as a business case that he/she can put forward to the board of directors in case it requires their approval. Once approved, this document stands as proof for initiating the project.

Appoint a Project Manager

The *Sponsor* or another relevant authority identifies a *Project Manager* for executing the project. Larger organisations identify the *Project Manager* from within or hire from outside.

The *Sponsor* or the respective authority creates a document called the Project Charter which is an extension of the Project Initiation Document and authorises the *Project Manager* to manage the project.

Kick-off the project

Once the *Project Manager* is identified, he/she goes ahead with kicking-off the project. This activity is part of the planning phase with which the project officially begins.

In many cases, preparing just the business case takes longer than the project. There may be many influencing factors that need to be considered even before building a business case. This is most commonly observed in government projects.

Many large or complex programs may be divided into several smaller projects. In such projects, which are part of a large program or project, initiation happens at different stages based on the planning of the program.

During creation of the Project Initiation Document, stakeholders of the project need to be identified. Once the project creation is confirmed, the *Sponsor* can identify a *Project Manager* for the project and assign work. The primary task of a *Project Manager*, once assigned to a project, is to identify the *Stakeholders* of that project.

Identification of Stakeholders

A *Stakeholder* is an individual or an organisation that has interests, which may be positively or negatively influenced by the performance of the project and/or who may exert influence over the project.

Identification of *Stakeholders* is an important activity since the *Project Manager* has to manage their expectations. The *Project Manager* has to understand their role in the project, the extent of their influence on the project and how—and to what extent—they are affected by the project.

The *Project Manager* has to build the list of *Stakeholders*, speak to them, understand their expectations or influence on the project and constantly try to manage their expectations. One might ask what happens in the case of a drive run by the government to tackle an epidemic. Do we speak to the entire country's population to understand their expectations? No. In such cases—where numerous *Stakeholders* of a similar type are involved—a sample of the population is picked up and its expectations are documented and managed.



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Stakeholders can be classified into two categories—primary and secondary.

Primary Stakeholders

Primary *Stakeholders* have a major interest in the success of a project because they are directly affected by the outcome. Customers and end-users are primary *Stakeholders*. Additionally, some project *Sponsors*, *Project Managers* and *Team Members* are also included in this category.

The project *Sponsor* is accountable for keeping the project on schedule. It is the *Sponsors'* responsibility to schedule regular meetings to review timelines, address complications that may arise and ensure that the *Project Manager* remains on the task. *Sponsors* allocate and supply resources and finances to fund the project. Success of a project is largely dependent on the project *Sponsor's* leadership and support. The leadership provided by the *Sponsor* helps identify cost overruns and provides alternatives in order to remain on budget.

Secondary Stakeholders

Secondary *Stakeholders* are those who may not have a formal contractual relationship to the project but can have a strong interest in what is going on regarding the project. They assist with administrative processes, financials and legalities. Proper communication between primary and secondary *Stakeholders* ensures that everyone is working towards the same goal and prevents breakdown within the project.

Stakeholders can be both, internal as well as external.

Internal Stakeholders

Project managers are internal *Stakeholders* because they are directly involved in developing the project. They have the authority to manage the project by handling the responsibility of work performance, organising and planning—effectively ensuring that all phases of the project are managed accurately and efficiently.

External Stakeholders

Vendors, suppliers and outside organisations are external *Stakeholders* because they supply the elements necessary for a project's success. They need to be updated regularly about the goals, milestones and deliverables.

Stakeholders can be classified into yet another category—direct or indirect *Stakeholders*.

Direct Stakeholders

Direct *Stakeholders* are those who are affected by the day-to-day activities of a project. *Team members* are direct *Stakeholders* as their workloads are scheduled around the project each working day.



Indirect Stakeholders

Indirect *Stakeholders* are those who are not affected by the day-to-day events during the execution of the project. Those not affected are customers and end users, because their concern is with the finished project. This includes the quality of merchandise, price, packaging and availability.

In the case of an event organised by a college, the directors, the principal and the actors/key participants would be the primary *Stakeholders*. The organisers and attendees can be considered as secondary/indirect *Stakeholders*.

Responsibilities of Key Stakeholders

Let us look at some of the responsibilities of the key *Stakeholders* in a project.

Sponsor

He/she is a primary *Stakeholder* and decision maker. His/her interests and influence will be the highest in the project. The *Sponsor* should have a clear understanding of the project and his/her expectations. He/she provides financial support and resources to the project. He/she also influences these significantly. He/she can influence indirectly through approvals or by bringing in a support structure to help the project team. The *Sponsor's* expectations need to be clearly recorded and communicated to all other *Stakeholders*. He/she is the one looking for the complete status of the project and it is responsibility of the *Project Manager* to provide the status as and when required. He/she is empowered to stop the project at any point of time. He/she is also the one who pays for the project and has to be kept updated about all that happens in the project all the time.

Project Manager

The *Project Manager* is responsible for managing and executing the project. He/she is authorised by the project *Sponsor* to execute the project and is provided with all support. The *Project Manager* is like a guide driving you in a desert; without him, the project may very well get lost. His/her decisions, direction and guidance are of utmost importance and have to be accepted by the project team. He/she owns the complete responsibility of the project. We will look at the *Project Manager's* roles and responsibilities throughout the learning process.

Project Team

The team which reports to the *Project Manager* directly and takes his/her advice and guidance is responsible for completing the assigned work. In many cases, the success or failure of the project directly affects them. This, for instance, is relevant with respect to their payroll and job security. Even the *Project Manager* is a member of the project team and could also be affected by the outcome of the project. Being one of the primary *Stakeholders*, the *Project Manager* has to hear them and understand their concerns about the project.

Customers

Customers are affected by the project directly or indirectly. They can provide inputs to the project team directly by keeping the *Project Manager* aware or through the assigned channel. When the



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Project Manager delivers a project to his organisation's customers/clients, they may influence the project directly.

End Users

The end users are the final receivers of the project or the product outcome. The *Project Manager* has to keep in mind the impact on the end users. Based on the type of project, end users may affect the success of the project.

Summary

In this segment, you learned that:

- the *Project Initiation Phase* is the conceptualisation of the project,
- projects can be initiated by individuals as well as organisations,
- the organisational demand for initiating projects can be classified into several categories,
- projects are initiated by following a series of steps,
- *Stakeholders* are the ones who are either direct or indirectly affected because of the project,
- there are various categories of *Stakeholders* depending on their level of involvement, and
- the key *Stakeholders* have various responsibilities, which have a bearing on the project outcome.



Identifying a Project

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1. Introduction
2. The Project Identification Process
3. The Steps in Project Identification
4. The Importance of Project Identification
5. The Sources for New Project Ideas
6. Summary

Objectives

Upon completion of this segment, you will be able to:

- Explain the project identification process
- List the steps in project identification
- Describe the importance of project identification
- List the sources for new project ideas

Introduction

In the previous segment, we learned about the *Project Initiation Phase* and identified the project *Stakeholders*. In this segment, we will focus on the process of identifying a project and the importance of this exercise.

Identification of a project is a critical step in the project management process. It is an important stage because it can affect the entire process including that of sustainability of the project after completion and transfer to the operational phase. This process is based on certain essential conditions, which may differ from project to project. Project ideas do not immediately take a concrete shape, therefore, projects should be identified from the perspective of need or demand, whether at the community or national levels.

The Project Identification Process

Project identification is a continuous and repetitive process that involves screening, documenting, validating, ranking and approving viable project ideas for an organisation. The systematic effort of screening the ideas generated through various sources and bringing out the most viable and feasible alternative project ideas is the domain of project identification. While examining the filtered project ideas for further approval, it is imperative to check for the availability of resources and potential skills to execute the project and make it realisable. It is futile to select such an idea for which potential skills are either unavailable and/or resources are insufficient for deployment. Thus, identifying a worthy project idea is not a simple mechanism but rather calls for a conscious and robust investigation on the part of the organisation.



Steps in Project Identification

Like any other category of ideas, project ideas do not immediately take a concrete shape. An idea is first born, it is under incubation for some time, and consequently, it begins to take a definite shape. This process of project identification can be broadly divided into four stages.

The conceptual stage

In the conceptual stage, a number of project ideas are generated through interaction amongst various groups of intellectually and entrepreneurially inclined people, in the organisation, who have good knowledge and concern about the field where contributions are sought.

In this context, one has to examine the potentialities of development and the problems, needs and aspirations of the people in the concerned area. For example, in an Aerodrome Project in the vicinity of a developing city, the problems that the people of that city will face—due to the impact of noise pollution, traffic and evacuation from the area to other non-developed areas—have to be considered.

The screening stage

In the screening stage, that is, the second stage, the project ideas generated above are screened and a preliminary exercise is conducted to weed out bad or unviable ideas. Therefore, it is understandable that all the given project ideas would not pass the screening and only a few will get shortlisted for further examination. Such shortlisted ideas will be considered while promoting pre-feasibility studies.

The identification stage

The third and the fourth stages can be termed as 'investment opportunity study'. The identification stage is a preliminary-level investigation, yet extensively conducted to analyse fundamental strengths of the project idea and to explore various opportunities offered by the project. It has a limited objective of providing planners with a choice of project alternatives from which they can make a selection.

The pre-feasibility stage

The pre-feasibility study focuses on answering the essential questions such as; should we proceed with the proposed project idea? What are the benefits and drawbacks of the proposed project idea? How much time will it take to conduct the preliminary study, that is, pre-feasibility study on the financial and other related economic aspects of project viability? In order to support those pre-feasibility studies, all the organisational efforts get directed towards compiling suitable answers to those questions. After gaining confidence through the answers so supplied, a pre-feasibility study will be commenced either by the internal sources or by outsourcing to consultants. Furthermore, upon the satisfaction of the pre-feasibility report, a detailed feasibility study can be conducted to document a detailed project report.



The Importance of Project Identification

Project identification helps solve major problems in the planning stage and addresses the challenge of meeting diverse needs, while setting clear objectives for the proposed project. Good projects act as catalysts for economic growth and development, which helps to streamline fruitful ideas through appropriate actions and achieve the desired results. This encourages creativity and innovation so as to optimise the resource utilisation that is available in the economy, namely manpower, capital and raw materials and so on.

Project identification helps the management in the following ways:

- identifies and analyses the purpose and scope of a project in the given situation of an organisation or an economy;
- encourages value addition to the existing wealth and capital of an organisation or economy, as the case may be, by raising more funds for project execution;
- encourages a multi-cultural environment and gives scope of social and cultural compatibility in the organisation as well as the economy;
- improves and develops the specific and general infrastructure of an organisation and the economy;
- determines the basic benchmarks for deciding on various types and forms of project ideas to materialise;
- helps to identify the size of the investment and indicates on the complexity of the project;
- helps to define the location and find appropriate technology to be used while executing the project;
- helps to formulate better marketing strategies to enhance profitability;
- identifies different constraints in the project environment and formulation;
- evaluates the potential resources to convert an idea into reality; and
- analyses the policies of the government and various legal restrictions.

Sources for New Project Ideas

Following are some of the sources from which new project ideas may emerge:

Performance of existing industries

The present trends and past experiences of industries provide a good indication about the health of a particular industry. Changing profitability structure and break-even analysis of industries offer adequate information about the financial health of various industrial sectors. This provides a superficial outlook on the industrial health and, thus, carefully forecasted trend lines help in determining the future prospects in emerging project proposals.

It is necessary to carefully analyse the changing phases of business cycles in which different industries play their respective roles from time to time. For example, the telecommunications sector—especially mobile communications—is growing tremendously in emerging markets such as India where the mobile market is poised to cross 409.9 million units by the end of 2017.

This ever growing market gives the investors a good opportunity to innovate cheap and trendy designer mobile instruments that can withstand rigorous utility. Chinese manufacturers are taking



advantage of this growing market by distributing mobile sets at moderate prices to attract the consumers.

Availability of raw materials

Availability of suitable resources is a must for any project to take off. Easy availability of good quality raw material at cheaper prices is a boon for any project. Identifying appropriate technology and resource base will help design a good project idea and also make use of those untapped resources for the betterment of the organisation as well as economic health. For example, Compressed Natural Gas (CNG) engines were innovated for the automobile sector as it was a feasible substitute for petrol. These new resources help the organisation gain an edge over its competitors and also help the economy boost its GDP.

Availability of skilled labour

Not all projects can be implemented universally. There are specific projects which need defined locations, specified environmental conditions and suitable manpower resources. For instance, cement and automobile manufacturing units, textile and pharmaceutical units, sugar factories and power plants are examples of projects that not only need good infrastructure but also the support of appropriate technology and availability of suitable manpower.

Mobility of manpower and also natural resources such as water and coal is a major challenge, which is why power projects are developed in the areas surrounded by water facilities and coal mines.

Import/export statistics

Statistics on international trade reveal the scope to evolve good projects. For example, fruits, handicrafts, cereals, pulses and drugs have equal opportunity to get exported as well as imported depending on the demand by the country from time to time. Therefore, a clear understanding of the routine and specific international transactions between various segments of the society will make project ideas clearer and help reveal potential areas to evolve good projects.

Usually any product having high imports can be thought of as a potential project proposal for indigenous development and vice versa. However, it may not be true in all cases. For example, 80% of the caprolactum (raw material used to manufacture Nylon filament yarn) gets imported in India. Therefore, SRF thought of developing the project in the early 1990s but was unable to do so due to the heavy costs involved.

Price trend

The price of a product is a major driving force in making any project take off. Fundamentally, every organisation will have a twin focus—profit maximisation and cost minimisation. From both the angles, it is the price that gets determined and thereby guides the project initiator on the functionality. If the general price level is found to be rising over the past few years and a specific item is having a steeper rise, it would indicate a huge gap in the demand-supply and can be thought of as a project opportunity.



Data from various sources

Various publications of the government, banks, financial institutions, consultancy organisations, manufacturer's associations, export promotion councils, research institutions and international agencies contain data and statistics which may indicate prospective ventures. A study of the balance sheets of existing companies is useful in knowing the sectors of industry that are performing well. The study of profitability, break-even level and Earnings Per Share (EPS) of various industries may indicate those industries where opportunities exist for new investments.

Summary

In this segment, you learned that:

- identification of a project is a critical step in the project management process,
- project identification is the systematic effort of screening and choosing the most viable project idea,
- the process of project identification can be broadly divided into four stages, and
- ideas for new projects may emerge from various sources.



The Project Charter

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1. Introduction
2. Elements of an Effective Project Charter
3. Major Sections of a Project Charter
4. Summary

Objectives

Upon completion of this segment, you will be able to:

- Explain the purpose of a project charter
- Identify the elements of an effective project charter
- List the major sections of a project charter

Introduction

In the previous segment, we learned about the importance of identifying a project. In this segment, we will focus on the importance and the various elements of a project charter.

A project charter is a document created to formally communicate the existence of a project. It is issued during the *Project Initiation Phase* and is looked upon as the beginning of the *Planning Phase* of the project.

The project charter helps the project manager to communicate his authority and explain to the project participants and stakeholders the need for the project, the people it involves, the estimated project duration and its cost, the resources required and the benefits that the organisation will accrue from successful completion of the project. Once created, the document is rarely amended.

Depending on an organisation's culture and management style, a charter may serve the same purpose as a business case. In a large organisation, the charter may run into several pages written up by mid-level management after a business case has been approved but before the project scope has been defined. In a small start-up, however, the charter might just be a few paragraphs—containing bulleted items—signed by the president of the corporation.

Elements of an Effective Project Charter

An effective project charter should answer the following questions:

- **Why?** Explains the goals and benefits of completing a project. A good project charter provides a description of tangible benefits to the organisation, including quantification of the potential range of benefits (optimistic–realistic–pessimistic).
- **What?** Describes the high-level scope of the initiative and the key requirements associated with the request (often in the form of critical success factors).
- **Who?** Provides a list of the primary person/organisation requesting the project (*Sponsor*), and the key people/organisations affected by the project (stakeholders). It is helpful to



describe how the people/organisations are affected (for example, suppliers, subject matter experts, end users and customers).

- When? Describes the timing-related requirements associated with the fulfilment of the project request (for example, the window of opportunity). In some cases, how the project request will be fulfilled is known, and the project charter provides key project milestones and target dates.
- How? Describes what is known about the project delivery process. In many cases, the project does not begin from a 'blank piece of paper'. The project may represent an enhancement to an existing product or system, or the implementation of a product/technology that has already been identified or purchased.
- How much? Establishes funding available to complete the project request (for example, the amount budgeted in the organisation's budget) and/or cost-related parameters associated with fulfilling the project need. In many cases, the project charter will define a rough order of magnitude-type estimate for completing the project (for example, +/- 50%).
- What else? An effective charter includes any additional information that is relevant to approving the project request and initiating the next step in the project delivery process (for example, the project assumptions or constraints). The project charter may also include an assessment of risks and issues associated with the project request.

Based on the fulfilment of the above requirements, following can be listed as the elements of an effective project charter:

- Background/description
- Goals and benefits
- High-level scope
- Key product requirements
- Critical success factors
- Proposed project delivery process
- Project sponsor and key stakeholders
- Cost estimates/targets
- Target dates/milestones
- Assumptions and constraints
- Known risk and issues
- Score/ranking

Major Sections of a Project Charter

The template of a project charter will typically have the following major sections:

- General information
- Project background
- Project objective
- Project scope
- Project constraints
- Project authority
- Roles and responsibilities
- Critical success factors
- Disaster recovery methodology
- Signatures



Summary

In this segment, you learned that:

- a project charter formally communicates the existence of a project as well as the *Project Manager's* authority,
- the effectiveness of a project charter can be determined by asking some key questions, and
- the project charter is in the form of a template and comprises various sections.



The Project Life Cycle

Table of Contents

1. Introduction
2. Phases of the Project Life Cycle
3. Summary

Objectives

Upon completion of this segment, you will be able to:

- Explain what project life cycle means
- Describe the various phases of the project life cycle

Introduction

In the previous segment, we learned about the purpose of creating a project charter. In this segment, we will focus on the concept of project life cycle and identifying its various phases.

The project life cycle refers to a series of activities that need to be performed to achieve the project goal. These activities are grouped into phases.

The project life cycle mainly consists of five phases, namely *Initiation*, *Planning*, *Execution*, *Monitoring and Control* and *Evaluation* (closure). Execution is one of the longest phases of the project life cycle and broadly covers *Execution*, *Monitoring and Control*. Each phase has its own sub group and associated processes.

Let us take a quick look at these phases.

Phases of the Project Life Cycle

Initiation

This phase involves analysing, examining and carefully making decisions to initiate a project based on the goals that need to be achieved. A project is formally started, named and defined at a broad level during this phase.

Planning

This phase involves creating a proper project plan—outlining the work that needs to be done—and getting approvals to begin the project. Some of the important activities that mark this phase are making WBS; developing schedules, milestone charts and GANTT charts; estimating and reserving resources; and planning dates and modes of communication with stakeholders based on milestones, deadlines and important deliveries.

Execution

This phase involves utilising resources and skill and performing actual work on the ground. A lot of project management tasks during this phase capture project metrics through tasks such as status



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meetings and project development updates, status reports and human resource development and performance reports.

Monitoring and Control

This phase mostly deals with measuring the project performance and progression with respect to the project management plan. It involves monitoring each and every activity of the project on the ground and taking appropriate measures to ensure that the project is heading in the right direction.

Closure

This phase involves formal closing of the project by documenting the outcome and recording it for future use. It includes a series of important tasks such as making the delivery, relieving resources, reward and recognition to the team members and formal termination of contractors in case they were employed on the project.

Summary

In this segment, you learned that:

- project life cycle refers to the activities that need to be performed to achieve the project goal, and
- *Initiation, Planning, Execution, Monitoring and Control* and *Evaluation* (closure) are the five phases of the project life cycle.



Selecting Projects

Table of Contents

1. Introduction
2. Project Portfolio Management (PPM)
3. Project/Process/Product Comparison Grid
4. Product Decision Tools
5. Summary

Objectives

Upon completion of this segment, you will be able to:

- Explain the importance of project selection
- Describe the steps to determine which projects will comprise a given project portfolio
- Explain the method used to gather high-level information about a company's initiatives
- Identify projects that should be included in the project portfolio using the project process/product comparison grid

Introduction

In the previous segment, we learned about the project life cycle and its various phases. In this segment, we will focus on the process of project selection and its various aspects.

Most organisations consider multiple projects—at the same time—as against restricting themselves to a single project. Such a practice is known variously as:

- Program Management,
- Multi-project Management,
- Portfolio Project Management (PPM), and
- Enterprise Project Management.

This list is, however, not all-inclusive; there are no doubt other descriptions for the set of projects that an organisation might consider at any given time. Using project selection models, an organisation may find that even though some sets of projects are necessary, it simply cannot consider all of them or take advantage of all the available opportunities.

Portfolio project management is one of the more widely-used descriptions of the mechanism for selecting sets of projects because of the widespread use of the term 'portfolio' as used in an investment portfolio in finance. As in financial decision-making where we must manage our portfolio of investments wisely, an organisation too must manage its portfolio of projects well if it is to be successful.

To understand this phenomenon better, let us consider the following analogy:



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What do we include in our investment portfolio?

We include some:

- risky assets,
- not so risky assets,
- stocks of possibly large companies,
- stocks of small companies,
- growth issues, as well as
- income issues.

Now think about what an organisation should include in its project portfolio. As in the case of an investment portfolio, an organisation will include—in its project portfolio—projects which:

- are short term,
- are long term,
- take it in entirely new directions,
- are incremental improvements to existing products or processes,
- result in entirely new products, and
- represent extensions or new applications of old products.

Because of the ever-growing popularity of project management and the fact that organisations are adopting a 'Management by Projects' approach, it has become very difficult to identify which projects will be adopted. In such a scenario, how do executives determine which projects are going to:

- help meet the overall goals of the organisation?
- fit into the strategic plan?
- be successful additions to the firm's portfolio of projects?

In this section, we will find the answers to these questions and will learn about Project Portfolio Management (PPM). We will also understand how project selection models help determine which projects will become part of the project portfolio.

Project Portfolio Management (PPM)

So, what is Project Portfolio Management?

According to the Project Management Institute, Project Portfolio Management refers to the selection and support of projects or program investments. Such investments are typically driven by an organisation's strategic plan and available resources. With PPM, an organisation increases its chances of success by aligning the available resources more accurately with its strategic plans.

The following box shows the three steps in compiling a project portfolio:

Gathering information on initiatives

A project portfolio begins with gathering information about an organisation's initiatives. Often, the management is not fully aware of the projects that are currently underway. Better yet, simply identifying all of the project initiatives often reveals that resources are over-allocated.



By taking the PPM approach, organisations can eliminate projects doomed for failure before they even begin. PPM can also help to see which projects overlap and may be better off consolidated into one manageable effort. The reverse of this is identifying larger projects that may be better managed if broken into several, separate projects.

In order for PPM to be successful, an organisation must have in place a strategic planning process that is consistently implemented throughout the organisation. Project management methodologies must also be consistently implemented (for example, progress reporting). In turn, the management must fully understand the reports that are presented, and, more importantly, have a good understanding of project management.

Collecting data on individual projects

Once the high-level information on the organisation's initiatives has been gathered, it is important to elaborate on the data for each project that is underway. Information such as resource skills (for example, programming language in an IT portfolio application) can help a company determine resource needs and allocation.

Selecting projects

Finally, upon the completion of data collection, the use of project selection models can be implemented to determine which projects will comprise the strategic portfolio.

Gathering high-level information about an organisation's initiatives

As indicated above, the first step in determining which projects will comprise the project portfolio is to gather high-level information about the organisation's current initiatives. If an organisation must meet its overall strategic goals, it needs to focus its efforts on projects that will help in meeting these goals.

We cannot be diverted into other areas and spread ourselves too thin where we do not have a great deal of expertise or a competitive advantage.

We cannot, for example, pick up a project simply because a salesperson feels we should, or an engineer thinks it would be an interesting one to work on. If our objective is, for example, to reduce emissions by 5% per year, we should not be adding the types of projects (efforts) that contribute additional pollutants that we cannot control.

If one of our strategic objectives is to maintain full employment levels, we should complement labour saving projects with ones which will find new work for our employees as a part of meeting our strategic goals.

PPM enables organisations to make such strategic decisions on the types of projects to take on.

Following is an example of an executive letter that an organisation can use to gather high-level information about its initiatives.

From: Robert Smith

To: Organisation Members



In order to ensure that our organisation focuses its effort on projects that will help in meeting our overall strategic goals, we have decided to implement Project Portfolio Management. To begin our efforts, we need to identify all of the work currently underway or proposed in our organisation. After completing this exercise, many organisations find that they have over-allocated resources by two to three times. Your input will help us to more effectively align resources with our strategic plans.

Please fill out the table below and submit your list of current and planned work to Sarah Olan by _____. I will also be available to answer any questions you may have regarding portfolio management.

Sincerely,

Robert Smith

Project Process/Product Comparison Grid

The project selection model adopted by a firm should generally take into consideration a project's strengths, its weaknesses, opportunities and threats in comparison to its benefits and resource requirements. There are two basic forms of project selection models, numeric and non-numeric. Projects must also be compared to how they complement or detract from other on-going efforts or initiatives in the organisation.

Many organisations will use both forms of selection modes to determine which projects should be added to their strategic portfolios. Before determining which selection model(s) will be used, an organisation must first, of course, identify its overall objectives and strategic plans.

The following box illustrates a grid that can be used for this purpose:

Project Comparison Grid

One useful step in narrowing down the identified projects that will be selected in developing the project portfolio is to identify the categories of projects and place each of the potential projects into the categorised table or grid.

In doing so, we will often find commonalities as well as differences in them. We will be able to visually obtain an estimate of the preliminary risk and learn where there are many conflicts on resource usage or over usage.

The first step in comparing the alternatives in an aggregate or a structured format is to construct a scale which shows the degree of process change on the vertical axis (minor to extensive) and the degree of product change on the horizontal axis (extensive to minor product change) for a project.

Competing projects are classified as:



Research and development (R&D) projects

Visionary endeavours oriented to developing new technologies or using technologies in entirely new ways. New knowledge generation is an example of a research and development effort.

Breakthrough projects

These projects involve new technologies for the firm, but not unknown technologies in general.

Platform projects

These projects represent new platforms of product offerings for the organisation. They spawn a new generation of product or service offerings. An example might include the development of the SUV platform in automobile production or development of the mini-van which took place some years ago.

Derivative projects

These projects offer new programs or services that are only incrementally different from previous offerings by the firm. For example, power side opening doors in a mini-van certainly extend the usefulness of the product and perhaps safety when young children are present, but do not result in an entirely new product or platform from which other products or services will be developed.

Once classified, projects are plotted according to size, the larger circles represent corresponding larger projects in terms of resource usage or project scope or length (duration). Different colours can be used to indicate different divisions involved as well.

These aggregate maps can be used to visualise the project portfolio at a glance and are a useful first step in making recommendations for a project mix. This classification process was first described by Kim Clark and Steven Wheelwright of Harvard University in 1992.

Project Decision Tools

Portfolio decision tools can also be used to evaluate projects as to whether or not they will become part of an organisation's portfolio.

Following are the major project decision tools:

Expected Commercial Value (ECV)

ECV uses decision tree-analysis to break projects into decision stages. It defines various project outcomes with their associated probabilities. Projects are then indexed and ranked in order to maximize profitability.



Scoring Model

Organisational decision makers rate projects (usually on a scale of 1-5 or 0-10) based on a series of questions that help to differentiate 'priority' projects. Scores are compiled and prioritised accordingly until all the resources are exhausted. Those projects that do not have enough resources to get the job done are eliminated from the project portfolio.

PMOffice

Some organisations develop or purchase their own tools to assist with project portfolio management. One such tool, recently acquired by IBM, is PMOffice. PMOffice is a real-time, web-centric and enterprise-scalable tool that helps organisations with portfolio management by:

- establishing total internal and external resource pools and capacities,
- balancing the supply and demand of workers,
- connecting to the broadest selection of global resource pools,
- aligning skilled resources with strategic engagements,
- capturing best practices and sharing project intellectual capital with all practitioners,
- tracking and controlling project time and expenses across multiple engagements,
- managing mission critical processes, and
- evaluating the performance of a particular project, team, program or business area.

Whichever tool or model an organisation uses to aid its Project Portfolio Management, it is important that the idea of a balanced portfolio be kept in mind. Ranking projects, prioritising and shuffling projects around until all resources are exhausted is not enough to determine which projects should be selected.

Some questions that should be kept in mind when trying to balance the portfolio include:

- Does the project align with the corporate strategies?
- What is the stability of the project's competitive advantage?
- What are the financial rewards with respect to investment?
- What are the costs and time involved?
- What are the risks associated with the project?
- What is the probability of success?

Summary

In this segment, you learned that:

- Project Portfolio Management refers to the selection and support of projects or program investments;
- a project portfolio begins with an assessment of all the organisation's initiatives;
- with project portfolio management, an organisation increases its chances for success by more accurately aligning available resources with their strategic plans;
- in order for Project Portfolio Management to be successful, an organisation must have in place a strategic planning process that is consistently implemented throughout the organisation;



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- one useful step in narrowing down the identified projects that will be selected in developing the project portfolio is to identify the categories of projects;
- other tools besides project selection models—that may assist with portfolio management—include ECV and the Scoring Model; and
- it is important to keep in mind that project portfolios should be balanced.



The Project Boundary

Table of Contents

1. Introduction
2. The Project Management Triangle
3. Creating a Project Boundary
4. Summary

Objectives

Upon completion of this segment, you will be able to:

- Explain the concept of a project boundary
- Describe the project management triangle
- Identify the characteristics of a good project boundary

Introduction

In the previous segment, we learned about the process of selecting projects and the various considerations that are taken into account while doing so. In this segment, we will focus on understanding the concept of a project boundary and its constituents.

Before beginning a project, a huge amount of effort has already been spent on choosing the right project. Individuals do a lot of thinking, speaking to friends, colleagues and meeting experts in that area before starting on a project. Big organisations spend a lot of money to understand the market for their projects. They rope in market analysts, strategists and subject matter experts to provide insight and analysis.

Does this mean that projects can run for an indefinite period of time?

Nothing could be farther from the truth!

Have you heard of Tejas—the ambitious Light Combat Aircraft (LCA) project of the Indian Air Force?

Although, initially, the work and expected result of the LCA were identified before the work was begun, many new changes, challenges and constraints kept on creeping into the project.

The magnitude of changes in this project are apparent from the fact that it was first sanctioned in 1983 at a cost of INR 71.16 billion and was initially envisaged to be completed in ten years. However, it has taken more than twenty years to complete, the project cost has shot up to INR 140.33 billion and it remains a work in-progress.

The impact of such a project is not difficult to imagine—increased cost, over-running schedule and the waning patience levels of the users waiting for the output of such a project. Depending on the type of project, there can be many other potentially undesirable effects.



Project Management

Let us take the example of a typical wedding. Assume that you have hired a wedding planner and one of his responsibilities is to serve food on the occasion. Now, assume that that planner does not get the approximate number of guests and the expected count of dishes, but, nevertheless, accepts to serve food to all the guests for a fixed amount. What could happen? Is the planner likely to manage any profit or is he/she headed for a loss?

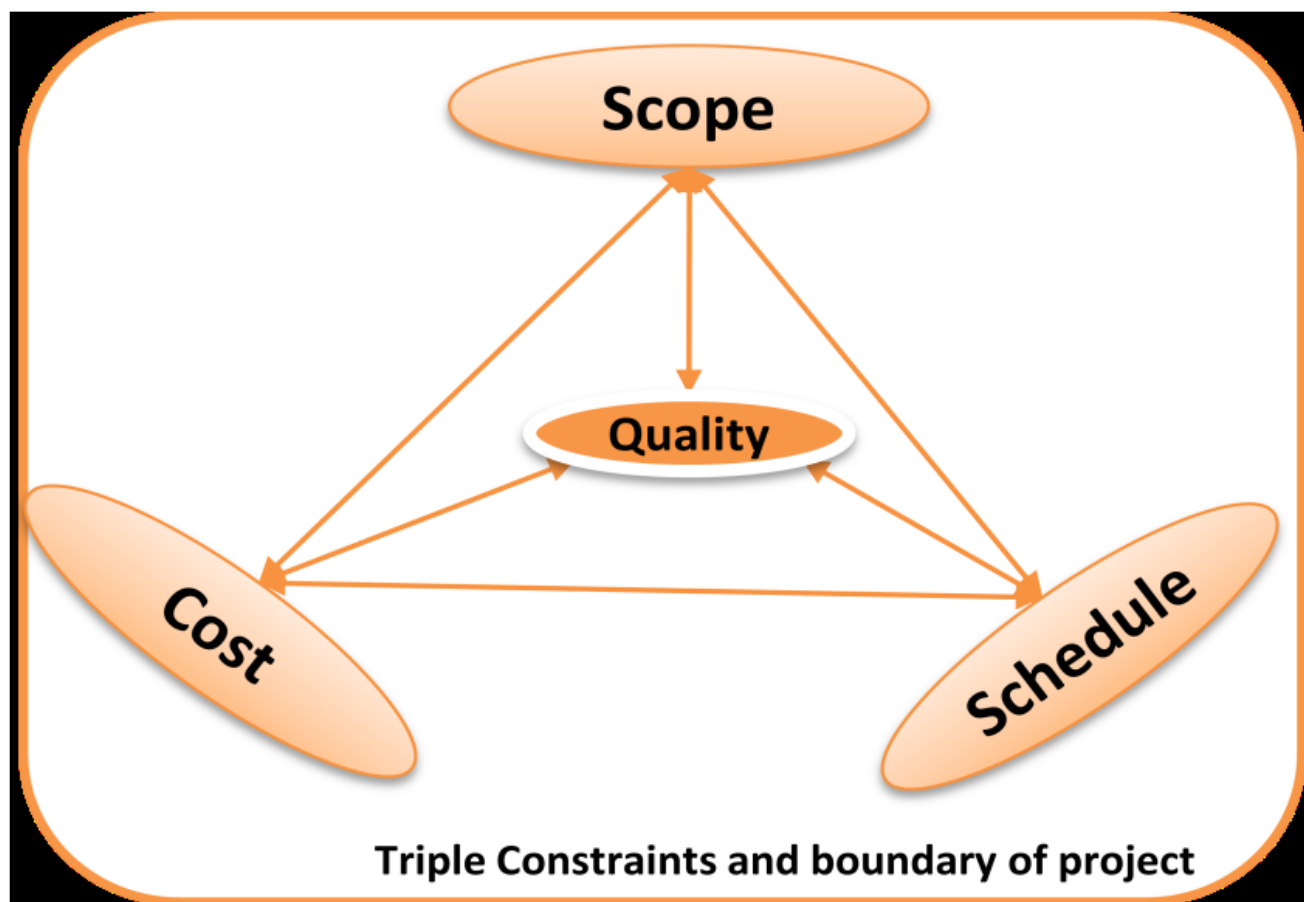
If the number of attendees is less than what was estimated, the planner could very well make a profit; although, he/she is likely to waste food. If the attendees are more than what was estimated, he/she would certainly incur losses.

The planner cannot come back and ask for more money as he/she had accepted the project without any boundary!

Similarly, the wedding cannot go on for an indefinite duration. The ceremonies must be completed by a specific time and within the planned budget.

A project boundary, therefore, means a clear definition of the problem domain and the functionalities/features or service that is being provided as part of the project undertaking. A good way to conceptualise the project boundary is to think of it as the interaction of the project *Sponsor's* needs and his/her stated requirements (not union or either one).

The Project Management Triangle





Project Management

The project management triangle helps us understand that the triple constraints form an equilateral triangle. An equilateral triangle is a triangle whose sides are of equal length. Any pull or push (influence) on one constraint will affect the equilibrium of the project.

Let us understand these triple constraints—scope, schedule and cost—in some detail.

Scope

The project requirements—which must be clear—that we discussed in the example of the wedding planner are termed as the *Scope* of the project. It can be defined properly when the requirements of the project have been stated clearly. Underlining the scope is an importance job of the *Project Manager*.

Setting the project scope boundary does not mean that the project requirements or specifications must be unchangeable or 'set in stone'. While you can change the scope boundary over a period of time, such changes—as explained earlier—will result in rising costs when you are in the midst of the execution of the project. The project *Sponsor* must be made aware of such changes and their impact on the cost and timeline of the project.

Time (Schedule)

The *Project Manager* cannot continue to execute the project indefinitely. As defined earlier, there must be a *time* limit. Time is another essential boundary of the project. The timeline of the project can be arrived at based on the requirements of the project as well as the experience and scientific techniques available.

Timelines can vary based on various factors such as the number of resources available, need for business or even legal issues. This has to be derived upfront during the initiation of the project. If it is not possible to estimate an accurate time frame, the expected duration within which a project has to be completed must be defined. This is where estimation of work, schedule and cost comes into focus.

Cost

Cost of the project is another boundary within which the *Project Manager* has to complete the project. Since there can be changes while executing the project, the *Project Manager* has to keep an eye on the cost and keep the *Sponsor* updated if there is any rise cost. If the *Project Manager* understands that there is an impact on the original planned cost—due to any reason—he should make the respective stakeholders aware and take appropriate decisions.

The expected outcome of the project must meet the goal of the project along with the *quality* expected by the stakeholders. Although quality is not a part of the project management triangle, it is the ultimate objective of every delivery.

Many project managers are under the notion that 'high quality comes with high cost', which—to some extent—is true. Using low quality resources to accomplish project deadlines does not ensure success of the overall project. However, high cost with little regard to the potential return on investment (ROI) will also not guarantee quality and overall project success. The Project Manager, therefore, has to do a careful balancing act in order to meet the stated project objectives.



In this way, *scope*, *time*, *cost* and *quality* are the key constraints of a project which form the project boundary. There are many factors that affect these and it is the responsibility of the Project Manager to ensure that project stays well within these boundaries.

Creating a Project Boundary

So, how do we create a proper project boundary or scope of the project?

In an ideal situation, all the project requirements must be frozen before starting a project. However, due to many practical constraints, this rarely happens. Even before the Business *Sponsor* gives a go-ahead to the *Project Manager* to start a project, there is a lot of work that needs to be done.

A thumb rule for arriving at an effective project boundary is to have a proper definition of requirements. These definitions—when questioned—should always lead to a yes/no response rather than a ‘maybe’ or ‘could be possible’.

A good project boundary:

- states the goals of the project in clear words,
- leaves no ambiguities when specifying requirements,
- clearly specifies the numbers wherever the expected outcome of the project can be quantified,
- clearly mentions the lower and higher ranges if there is a range in the expected output,
- clearly mentions when to expect clarity, in case there are uncertainties in the requirements even before the start of the project, and
- clearly mentions when to expect the requirements and how they could affect the project, in case there is any dependence on a third party.

A project boundary that fulfils the above criteria provides structure to the project and helps the *Project Manager* in running the project effectively.

Summary

In this segment, you learned that:

- the project management triangle comprises the triple constraints of scope, schedule and cost, with quality as the ultimate deliverable,
- the triple constraints provide structure to the project and define the project boundary,
- proper definition of requirements is a must for an effective project boundary.



The Project Kick-Off

Table of Contents

1. Introduction
2. What is a Project Kick-off?
3. Essentials of an Effective Project Kick-off Meeting
4. Summary

Objectives

Upon completion of this segment, you will be able to:

- Explain the significance of a project kick-off meeting
- Identify the essentials of an effective project kick-off

Introduction

In the previous segment, we learned about the project management triangle and the concept of a project boundary. In this segment, we will focus on the importance of an effective project kick-off.

The project kick-off meeting is a significant event in the project life cycle as it formally begins a project and is the *Project Manager's* chance to set the overall tone of the project!

The kick off meeting is called by the *Project Manager* who has the approval from the *Project Sponsor* to manage this project. The project charter gives him responsibility to call this meeting to kick-off the project.

The *Sponsor* approves the budget for the *Project Manager* to get the required resources. The *Project Manager* has to work with the human resource team to get the right resources required for the project. If new machines or accessories are required, he/she then needs to work with procurement and finance team to get them on time. While this process is on, the formal project kick-off meeting is held with the human resources of the project to set expectations and begin work on the ground.

The *Project Manager* can hold another kick-off meeting with his support team which could involve human resource (HR), finance, infrastructure, transport leads and so on. This is to apprise them of the project that is due to start and the kind of support that will be expected of them.

If there are time constraints and deadlines in the project, they can be introduced in these discussions and support can be sought. It could be things such as the transport team needing to arrange for extra vehicles exclusively for this team to work over the weekends.

It has been understood till now, that the project can be initiated by the project/business *Sponsor*. The *Sponsor* puts his money and has a vested interest in the project. Either he/she has been authorised to do the ground work for the project and give confirmation or himself/herself has the capability to do so.



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Once the project has been through the analysis and has been officially approved for kick off, the *Sponsor* chooses the *Project Manager*. In many cases, the *Project Manager* may already have been chosen and is part of the initial analysis of the project.

However, official communication from the *Sponsor* to all *Stakeholders* introducing the *Project Manager* is necessary. An officially created and 'approved-by-the-Sponsor' project charter is the document that acts as confirmation for the *Project Manager* to take care of the project.

The project charter mentions the role, responsibilities and the extent of decision-making authority given to the *Project Manager*.

The kick-off meeting is one of the important meetings which the *Project Manager* has to hold with his team and other *Stakeholders*. This project kick-off meeting is an opportunity for a *Project Manager* to energise his team. During this meeting, the *Project Manager* can establish a common goal and start understanding each individual. As a *Project Manager* needs to closely deal with individuals during the period of the project progress, this helps him understand his team.

Essentials of a Successful Project Kick-Off

Although a project kick-off meeting appears to be a simple meeting with all the stakeholders of the project, a successful project kick-off meeting requires proper planning.

The project kick-off meeting is the only meeting where all *Stakeholders* will attend, and it is the only platform which enables all *Stakeholders* to re-iterate their expectations from the project in the presence of other *Stakeholders*.

Following are some of the important preparation points for a successful project kick-off meeting.

The agenda

A strong and clear agenda is a must for a project kick-off meeting. If you have no clue of what the agenda should be, ask your experienced subordinates or read up on the agenda used for earlier kick-off meetings by others.

The agenda usually includes the purpose of the project, deliverables and goals, key factors necessary for the success of the project, communication plan and the project plan.

It must be made sure—in advance of the project kick-off meeting—that the agenda is circulated to all the participants. This way, all the participants are aware of the structure of the meeting and what is to be achieved at the end of the meeting.

Getting started

When the meeting starts, the *Project Manager* should take charge of the meeting. All the participants should be welcomed and a round of self-introduction should take place. Although the meeting agenda is shared with the participants, they should be briefly taken through the agenda giving a brief introduction to each item on it.

More attention should be paid towards introducing the project roles and in emphasising the reasons why the team members were assigned the respective roles. The *Team Members* should be provided an insight into the project and organisation structure. If there are people playing stretched



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roles, it should be acknowledged. It is not required to go into details, while performing the above activities as the purpose of this meeting is only to get everyone on to the same platform.

Presentation

Once the tone is set, the agenda should be presented in a structured manner. The *Project Manager* should talk about the project assumptions and how the project plan was developed. He/she should present the reasoning behind the plan and convey the message that suggestions will be welcome when the project progresses.

Each task in the project plan should be discussed and elaborated, as required. It must be emphasised that the project plan and the schedule are still at the initial stage and everybody's assistance is more than welcome to make it complete. This is also the stage to identify and acknowledge any potential bottlenecks or challenging tasks in the project schedule.

Setting the expectations

A convenient time must be chosen to hold regular meetings to talk about the project progress. It should be emphasised that everyone needs to participate in these regular meetings. Teamwork is one of the most important expectations to be set. The importance of teamwork must be reiterated and some teambuilding activities should be planned just after the project kick-off.

It is recommended to talk about the time sensitive nature of the project and how the leaves will be granted during the project period. If the project requires working long hours, letting the team members know of this in advance and showing them how you can help them to maintain the work-life balance is also a good strategy.

Communication

Communication is one of the critical aspects of a project. Therefore, the project kick-off meeting should emphasise more on the communication plan for the project. This usually includes the meetings and escalation paths. Following are some of the meetings that take place during the project life cycle:

- Weekly status meeting
- Project plan updates
- Task and activity planning sessions
- Management updates
- Daily stand-up meeting before beginning work (in some cases)
- E-mail etiquettes

In addition, we can emphasise on the other communication channels such as e-mail communications, forums and so on.

Closure of the meeting

Finally, at the end of the kick-off meeting, a Q&A session should be opened to allow the team members to freely express themselves. The minutes of this meeting should be documented and shared with all the attendees.



Project Management

In this way, project kick-off is an important aspect of the project management life cycle that helps the *Project Manager* set the tone for the entire project and clarifies expectations and from the *Team Members*.

Summary

In this segment, you learned that:

- project kick-off is a meeting that formally initiates a project,
- it helps the *Project Manager* communicate his/her expectations to the team members and sets the tone for the entire project, and
- the preparation points for a successful project kick-off include setting the agenda, getting started, making the presentation, setting the expectations, preparing a communication plan and bringing the kick-off to a closure.



Difference between Projects and Programmes

Table of Contents

1. Introduction
2. What is a Project?
3. What is a Programme?
4. The Essential Difference
5. Summary

Objectives

Upon completion of this segment, you will be able to:

- Describes what constitutes a project
- Describes what constitutes a programme
- Identify what differentiates a project from a programme

Introduction

In the previous segment, we learned about the importance of the project kick-off meeting. In this segment, we will focus on what constitutes a project and what its chief features are. We will also define what a program is and how it differs from a project.

What is a Project?

With the boom in Information Technology over last two decades, the word 'Project' has become a buzzword that is commonly used across the globe. It is used by everyone, right from school children to businessmen and scientists. Students in higher grades often discuss the topic of their project, which needs to be submitted at the end of the semester. Similarly the word project is often heard in many of the conversations among friends, business dealings, during management discussions and so on.

Is it really a new word?

Projects have been in existence since time immemorial. Maybe the term 'project' was not used in the past as widely as it is used now. The building of temples, aeroplanes, trains, and rockets were all projects.

Even in our day-to-day life, we have examples of projects such as renovating our house, cleaning our house after long time, digitising our paper photos and so on.

What, then, is essentially a project?

A project is any sequence of tasks performed to achieve a certain desired outcome by utilising resources for a certain period.

In case of a satellite launch, the outcome is to send it to the space and effectively utilise it for the specified purpose. In the case of renovating a house with new paint, the purpose is to make it look



Project Management

better and beautiful. Students working on their school project, are demonstrating in a practical form, what they have learned in the class. Thus, every project has a certain output and that is its expected outcome or goal.

Tasks are performed in a logically sequenced way so that the output of each task becomes the input for the next task and makes it easier to complete the project. Resources are required to perform any task in a project. These can be either human or mechanical.

Let us take the case of a student working on a project for his school. The project may be about writing an article for his school magazine. The goal of the project is to prove the student's knowledge or his ability to share his thoughts in a written form.

Following will be the sequence in which this project will be structured:

- a) Identification of an appropriate topic
- b) Studying and understanding the chosen topic
- c) Preparation of a draft
- d) Review of the draft
- e) Publishing the article

The above set of tasks could look simpler and can be performed in one go. Based on the volume of work and expected outcome, the above-mentioned tasks can be broken down as required. Breaking down tasks makes it easier to work and understand the position of the project.

In addition to this, all projects are time bound. If any activity is performed without a deadline, it cannot be called a project. Every project needs to have proper start and end time.

What are the characteristics of such projects?

Every such project that meets the above-mentioned requirements:

- is unique;
- is time bound;
- is a device for change;
- has a specific aim or goal;
- results in something getting delivered;
- is the responsibility of a single person or a group;
- involves cost, resources and time; and
- uses a wide variety of resources and skills.

The above characteristics are self-explanatory.

Every project is born with specific goals and seeks to be a catalyst for change. Its purpose is to bring in certain changes in the way things are currently done. It is time bound, in the sense that it has definite start and end date. Any activity that is not time-bound and continues endlessly, or without a specific end date, is called a service and not a project.

All projects have certain tasks to be performed, which are in alignment with the end goal. These are performed by either an individual or a group. This group could be a company, a set of friends, an organization or a society.

Finally, projects need money! They involve costs and resources which could either be human or mechanical, requiring skills and consuming time.



What is a Programme?

Building a satellite is an enormous effort. Although the components in a satellite appear to be small, building them requires precision and careful attention to detail. Each component needs to work properly and in perfect coordination with the other components. Each component has to provide and receive data and needs to have a smooth interface with the other. Chances of errors have to be minimal and are always planned to be zero.

Building each component itself is a project. Each project has its own expected outcome and has to feed the other at the right time. In such an environment, many project managers work on each component.

Let us take another example of constructing a dam. Can this be executed by only one person in charge?

It involves various activities, each one of them being a huge task. There are various important tasks like identifying the depth of the river and the quantity of water in different seasons, identifying the population of villages around the region before relocating them, acquiring land for the villagers and calling tenders for different activities. All these are executed as individual projects or phases of the bigger project of dam construction. These phases can have multiple projects within them, which are known as sub-projects or projects alone.

In this way, *when multiple projects are executed in a phased manner to achieve a larger goal, they constitute a programme.*

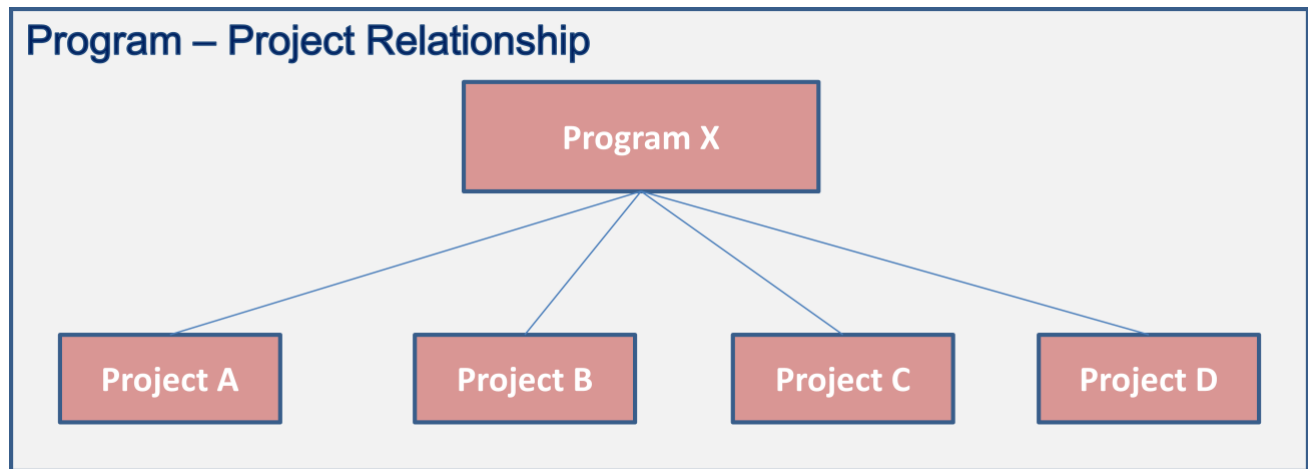
A programme is run by a *Programme Manager*. Many *Project Managers* report to a *Programme Manager*. *Project Managers* work synchronously to ensure that the complete satellite is built and launched into space in a time-bound manner or that the complete dam is built.

Programme Manager has the complete responsibility of overseeing and ensuring that the programme is completed as required. A programme is larger than a project and has a larger scope of work to perform and achieve.

The benefits that can be achieved by a programme are likely to be much higher than those achieved by individual projects. Generally, in small organisations, programmes are designed to deliver the organisational goals or strategy. In well-executed bigger organisations, each programme may seek to achieve bigger targets set by the organisation. Another example of a programme is construction of roads across cities, states or countries.



The Essential Difference



Projects and programmes, though inter-related, are different in many ways. Projects can be individual or part of a larger programme. Projects are executed by project managers whereas programmes are executed by programme managers. Outcomes expected from programmes are far higher/larger compared to projects. Responsibilities of programme managers are higher than those of project managers. Table 1 highlights the most notable differences between a project and a programme.

Table 1: Differences between a project and a programme.

Items	Project	Programme
Size	Small	Large
Responsibility	Project Manager	Programme Manager
Outcome	Low compared to a Programme	Very high
Subset	Projects are subsets of Programmes and lead to a common goal	Programmes contain multiple related projects

Summary

In this segment, you learned that:

- a project is any sequence of tasks performed to achieve a certain desired outcome by utilising resources for a certain period;
- when multiple projects are executed in a phased manner to achieve a larger goal, they constitute a programme; and
- projects and programmes, though inter-related, differ on many parameters such as size, responsibility and outcome.



Project Management and Programme Management

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1. Introduction
2. Project Management
3. Programme Management
4. Difference between project management and programme management
5. Summary

Objectives

Upon completion of this segment, you will be able to:

- Describe what constitutes project management
- Describe what constitutes programme management
- Identify what differentiates project management from programme management

Introduction

In the previous segment, we learned about what distinguishes a project from a programme. In this segment, we will understand the concepts of Project Management and Programme Management, respectively. We will also identify the features that distinguish Project Management from Programme Management.

Project Management

For a project to be successfully completed, it needs utmost attention and focus on details. The project needs a methodical approach to initiation, planning, execution, making on the field corrections and a closure.

Project management is an art and science to guide a project—from start to finish—for success.

It is an art because it is executed by humans and it can be executed in different ways. Each *Project Manager* has his own way of handling a project and its resources. Few *Project Managers* may follow processes as defined by their organisation. Many *Project Managers* take decisions instantaneously based on their gut feeling and experience. Even their working styles often vary.

Although the processes remain the same, there are various ways of executing a project depending on many factors such as the business domain, geographical location, culture, importance of the project, impact of the project and several other factors.

Project management is also a science because it uses many scientific tools and techniques. There are various recognised institutes, which define project management in different ways. At the end, it all boils down to the grouping of processes for the project life cycle.

Project Management Institute (PMI)—the leading body of professional project management professionals—defines project management as: *‘Project management is the application of knowledge, skills and techniques to execute projects effectively and efficiently. It is strategic*



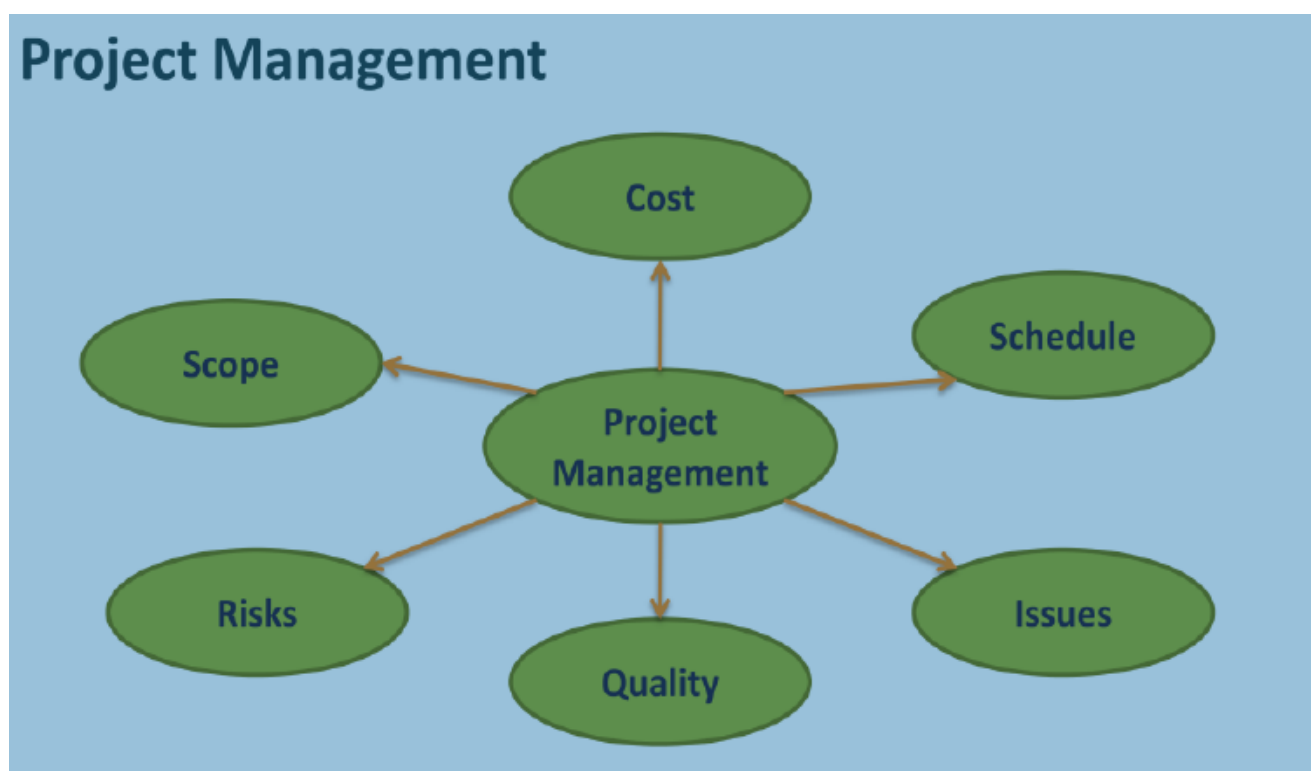
Project Management

competency for organisations, enabling them to tie project results to business goals and thus compete better in the market’.

Project management is all about how a project can be successfully managed to achieve its stated goals.

‘How to manage a project’ is defined by different organisations in different ways. Organisations set or define processes which are suitable for it based on their needs and relevance. Still, there are many organisations which may not have their own standards and processes but follow either the industry-defined standards or execute projects based on the skills of their project managers alone.

Project management involves managing key constraints of a project such as scope, schedule, cost and quality along with risks and issues to ensure it meets its goal.



Scope

It is the project's requirement that has to be completed by the end of the project.

Cost

It refers to the financial resources required to complete the project.

Schedule

It refers to the time required to complete the project.

Risks

These are unforeseen or foreseeable events that need to be managed with respect to the project.



Issues

These refer to events that can bring the progress of the project to a standstill.

Quality

It is the ability of the outcome of the project to satisfy its stated needs.

To summarise, there are many sides to project management and this is what makes it so interesting, challenging and demanding. This is a discipline that requires a great deal of time, skill and finesse. *Project Managers* are expected to take an uncertain challenge and make a certain promise to deliver. They are also expected to do this within a specified time and within a limited budget.

Programme Management

While projects are temporary in nature and have a definite start and end with goals tied to them, a program is a combination of multiple projects. The goals of programmes are bigger than those of projects and have a higher impact in terms of financial, social or any other need of an organisation.

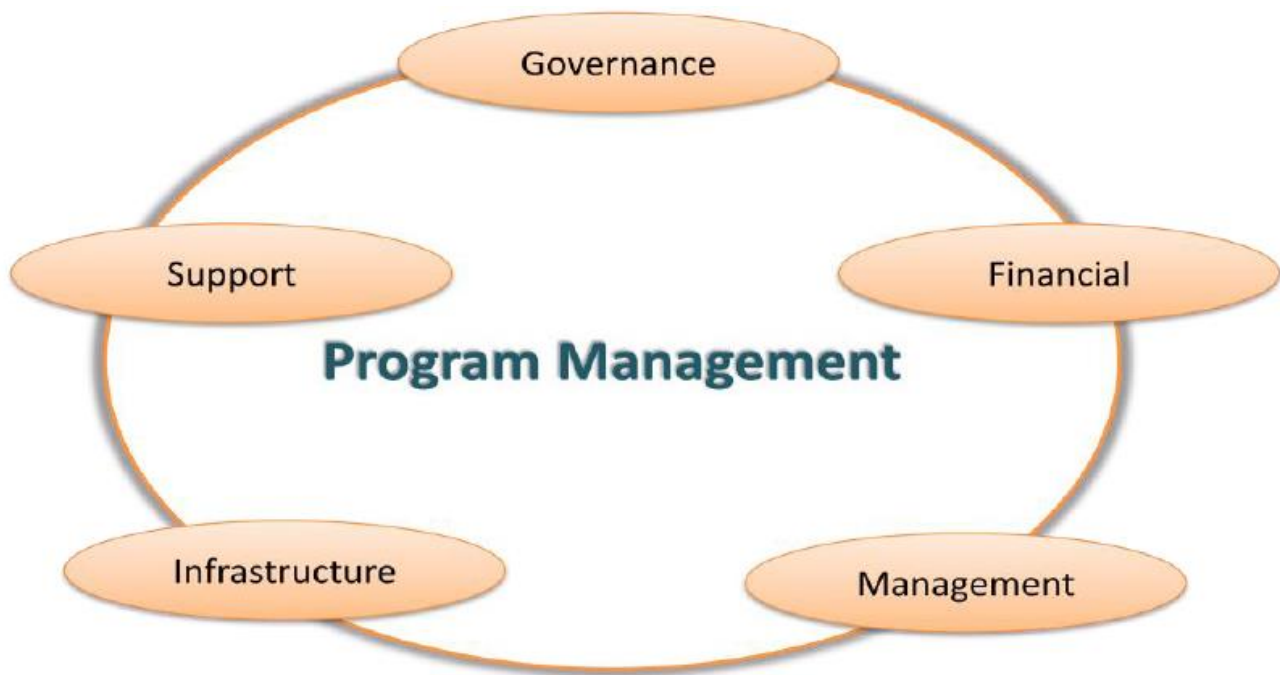
Programme management is the process of managing several related projects to achieve the common goal of all those projects. Typically, in a program, there is a need to identify and manage cross-project dependencies, risks, issues, requirements, design or solutions.

The *Programme Manager* has an overview of the purpose and status of all projects in a programme and can use this insight to support project-level activity to ensure that the overall program goals are met. The *Programme Manager* also enjoys a decision-making capacity that is not achievable at the project level.

The *Programme Manager* also has a programme perspective and, when required, can act as a sounding board for ideas and approaches for solving project issues that have an impact on the programme. The *Programme Manager* is also typically well placed in the reporting hierarchy to actively seek out information from *Project Managers*.

The main aspects of programme management are:

- Governance
- Management
- Financial Management
- Infrastructure
- Planning
- Support



Programme governance

It ensures the creation of, both, the structure and practices to guide the programme and provide senior-level leadership, oversight and control.

Management

It refers to managing and administering both projects and programs.

Financial management

It refers to the implementation of specific financial practices and controls.

Infrastructure

It refers to managing separate offices or support systems to provide support to projects.

Planning

The programme plan is not a traditional plan; it is an involved process. Drawing the plans and targeting their achievement through proper governance, management and support are all part of the planning process.

Support

It involves providing support to the programme through project management office.

In a nutshell, programme management is planning, organising, directing and controlling organisational resources to achieve the desired goal by breaking up a huge piece of work into many projects.



Difference between Project Management and Programme Management

The difference between project management and programme management can be understood with the help of the following table:

Item	Project Management	Programme Management
Responsibility of	Project Manager	Programme Manager
Hierarchy	Project Manager Reports to the Programme Manager	Programme Manager Reports to the Organisational Lead or the Board
Scope of work	Limited to Project	Bigger and Requires Projects to be Successful
Impact of work	Small Compared to that of a Programme	Very High Compared to that of Projects

Summary

In this segment, you learned that:

- project management is an art and a science to guide a project—from start to finish—for success,
- programme management is the process of managing several related projects to achieve the common goal of all those projects, and
- project management differs from programme management on various parameter including responsibility, hierarchy, scope of work and impact of work.



Portfolio Management

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1. Introduction
2. Portfolio Management
3. Summary

Objectives

Upon completion of this segment, you will be able to:

- Explain what is a portfolio
- Describe what is portfolio management

Introduction

In the previous segment, we learned about the project and programme management and the features that distinguish them from each other. In this segment, we will focus on understanding the concept of portfolio management.

A portfolio is a collection of programs or products put together. Based on whether the portfolio contains a list of products, projects or services, it can be called a product portfolio, project portfolio or a service portfolio.

Portfolio Management

Portfolio management refers to the process of analysing the strengths, weaknesses, opportunities and threats for performing a wide range of activities related to the one's portfolio for maximising the returns at a given risk.

Unlike project or program management, which involve directly managing the available resources and ensuring that the stated goals are achieved, portfolio management involves playing around the list and performing a wide range of activities related to risks, strengths, weaknesses and others to ensure maximum returns.

The objectives of portfolio management and the role of its manager vary from one industry to another. In an investment service, a portfolio manager may need to juggle with the list of mutual, equity, bonds or other funds to have maximum returns with the minimum risk. A portfolio manager may need to take decisions about investment mix and policy, matching investment to objectives and asset allocation for individuals and institutions.

In the case of a globally branded coffee shop, the regional portfolio manager may need to juggle with menu items in the catalogue to ensure maximum sale and return of consumers. He may have to look into tastes of the major consumers in the region and accordingly change items in the menu for sale.

A portfolio manager needs to make decisions on the collection of items in his portfolio based on various inputs and goals. He needs to constantly juggle the collection to ensure that the investment goals of the organisation on his portfolio are met.



Summary

In this segment, you learned that:

- a portfolio is a collection of programs or products put together,
- portfolio management is the process of analysing the strengths, weaknesses, opportunities and threats for performing a wide range of activities related to the one's portfolio for maximising the returns at a given risk.